

Answers_TIME & DISTANCE

1. **30 m/s;**

$$\text{Speed} = (108 \times \frac{5}{18}) \text{ m/sec} = 30 \text{ m/s}$$

2. **40 m/s;**

$$\text{Speed} = (144 \times \frac{5}{18}) \text{ m/sec} = 40 \text{ m/s}$$

3. **117km/hr;**

Distance = 650 m, Time = 20 seconds, Speed = ?

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{650}{20} \times \frac{18}{5} = 117 \text{ km/hr}$$

4. **54km/hr;**

Distance = 300 m, Time = 20 seconds, Speed = ?

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{300}{20} \times \frac{18}{5} = 54 \text{ km/hr}$$

5. **$1\frac{1}{5}$ km/hr;**

Distance = 400 m, Time = 20 minutes = $20 \times 60 = 1200$ seconds, Speed = ?

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{400}{1200} \times \frac{18}{5} = 1\frac{1}{5} \text{ km/hr}$$

6. **$2\frac{1}{10}$ km/hr ;**

Distance = 420 m, Time = 12 minutes = $12 \times 60 = 720$ seconds, Speed = ?

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{420}{720} \times \frac{18}{5} = 2\frac{1}{10} \text{ km/hr}$$

7. **5 : 9;**

$$\text{Reqd ratio} = 80 \text{ km/hr} : 40 \times \frac{18}{5} \text{ km / hr}$$

$$= 80 : 144 = 5 : 9$$

8. **10 : 27;**

$$\text{Reqd. ratio} = 120 \text{ km/hr} : 90 \text{ m/sec} \times \frac{18}{5}$$

$$= 120 : 324 = 10 : 27$$

9. **21 km/hr;**

5 revolutions are made in 3 seconds.

$\therefore$ 1 revolution will be made in $\frac{3}{5}$ seconds.

Distance covered in one revolution = circumference of the wheel i.e $3\frac{1}{2}$ meter = $\frac{7}{2}$ meter

$$\therefore \text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{7 \times 5}{2 \times 3} \times \frac{18}{5} = 21 \text{ km/hr}$$

10. **54km/hr;**

Distance covered in one revolution = circumference of the wheel i.e 12 meter

$$\therefore \text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{12 \times 5}{4} \times \frac{18}{5} = 54 \text{ km/hr}$$

11. **700m;**

Speed = 7km/hr , Time = 6 minutes = $6 \times 60 = 360$ seconds, Distance = ?

$$\text{Distance} = \text{Speed} \times \text{Time} = 7 \times \frac{5}{18} \times 360 = 700 \text{ m}$$

12. **1750m;**

Speed = 35km/hr , Time = 3 minutes = $3 \times 60 = 180$ seconds, Distance = ?

$$\text{Distance} = \text{Speed} \times \text{Time} = 35 \times \frac{5}{18} \times 180 = 1750 \text{ m}$$

13. **1 hr. 50 min;**

$$W + R = 1 \text{ hr } 20 \text{ min} = 80 \text{ min}$$

$$\therefore R + R = 80 - 30 = 50 \text{ min}$$

$$\therefore R = \frac{50}{2} = 25 \text{ min}$$

$$\Rightarrow W + R - R = 80 - 25 = 55 \text{ min} \quad \Rightarrow W = 55 \text{ min}$$

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$$\Rightarrow 2W = 55 \times 2 = 110 \text{ min} \quad = 1 \text{ hr. } 50 \text{ min}$$

14. **40 km/hr.;**

$$D_1 = 100 \text{ Km}, D_2 = 120 \text{ Km and } D_3 = 60 \text{ Km},$$

$$S_1 = 50 \text{ Km/hr}, S_2 = 60 \text{ Km/hr and } S_3 = 20 \text{ km/hr.}$$

$$\therefore T_1 = \frac{\text{Distance}}{\text{Speed}} = \frac{100}{50} = 2 \text{ hr},$$

$$T_2 = \frac{\text{Distance}}{\text{Speed}} = \frac{120}{60} = 2 \text{ hr. and}$$

$$T_3 = \frac{\text{Distance}}{\text{Speed}} = \frac{60}{20} = 3 \text{ hr.}$$

$$\therefore \text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{280}{7} = 40 \text{ km/hr.}$$

15. **40 min;**

$$W + R = 1\frac{5}{6} \text{ hrs} = 1 \text{ hr } 50 \text{ min}$$

$$\therefore W + W = 1\frac{5}{6} + 1\frac{1}{6} = 3 \text{ hrs}$$

$$\therefore W = \frac{3}{2} = 1 \text{ hr } 30 \text{ min}$$

$$\Rightarrow W + R - W = 1 \text{ hr } 50 \text{ min} - 1 \text{ hr } 30 \text{ min}$$

$$\Rightarrow R = 20 \text{ min}$$

$$\therefore 2R = 20 \times 2 = 40 \text{ min}$$

16. **30 km/hr.;**

$$D_1 = 120 \text{ Km}, D_2 = 90 \text{ Km and } D_3 = 60 \text{ Km},$$

$$S_1 = 40 \text{ Km/hr}, S_2 = 30 \text{ Km/hr}, S_3 = 20 \text{ km/hr.}$$

$$\therefore T_1 = \frac{\text{Distance}}{\text{Speed}} = \frac{120}{40} = 3 \text{ hr.}$$

$$T_2 = \frac{\text{Distance}}{\text{Speed}} = \frac{90}{30} = 3 \text{ hr.}$$

$$T_3 = \frac{\text{Distance}}{\text{Speed}} = \frac{60}{20} = 3 \text{ hr.}$$

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{270}{9} = 30 \text{ km/hr.}$$

17. **30 km/hr.;**

$$\text{Distance covered by car} = 3000 \text{ km, Time} = 50 \text{ hrs.}$$

$$\text{Speed of a car} = \frac{\text{Distance}}{\text{Time}} = \frac{3000}{50} = 60 \text{ km/hr.}$$

$$\therefore \text{Speed of a bus} = \frac{1}{2} \times 60 = 30 \text{ km/hr.}$$

18. **240 km.;**

$$\text{Distance covered on bike} = 600 \text{ km, Time} = 5 \text{ hrs.}$$

$$\text{Speed of the bike} = \frac{\text{Distance}}{\text{Time}} = \frac{600}{5} = 120 \text{ km/hr.}$$

$$\therefore \text{Speed of the bicycle} = \frac{2}{3} \times 120 = 80 \text{ km/hr.}$$

$$\therefore \text{Distance covered on bicycle} = \text{Speed} \times \text{Time} = 80 \times 3 = 240 \text{ km.}$$

19. **50 km/hr.;**

$$\text{Distance} = 5500 \text{ km, Time} = 110 \text{ hrs.}$$

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{5500}{110} = 50 \text{ km/hr.}$$

20. **105 km.;**

$$\text{Distance covered by train} = 3600 \text{ km, Time} = 4 \text{ hrs.}$$

$$\therefore \text{Speed of a train} = \frac{3600}{4} = 900 \text{ km/hr.}$$

$$\therefore \text{Average speed of a bus} = \frac{1}{6} \times 900 = 150 \text{ km/hr.}$$

$$\text{Time taken by bus} = 42 \text{ minutes} = \frac{42}{60} = \frac{7}{10} \text{ hrs.}$$

$$\therefore \text{Distance covered by bus} = \text{Speed} \times \text{Time}$$

$$= 150 \times \frac{7}{10} = 105 \text{ km.}$$

21. **36 mins.;**

$$\text{Manmeet's speed} = 60 \text{ km/hr.}$$

$$\text{Time taken by Manmeet} = 4 \text{ hrs.}$$

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∴ Distance covered by Manmeet = $60 \times 4 = 240$ km.

Given, Distance covered by Shivani = Distance covered by Manmeet = 240 km.

And time taken by Shivani = 3 hrs.

∴ Shivani's speed = $\frac{\text{Distance}}{\text{Time}} = \frac{240}{3} = 80$ km/hr.

Now, the average speed of Manmeet and Shivani increases by 20 km/hr.

New speed of Manmeet = $60 + 20 = 80$ km/hr.

New speed of Shivani = $80 + 20 = 100$ km/hr.

∴ Time taken by Manmeet = $\frac{240}{80} = 3$ hrs. and

Time taken by Shivani = $\frac{240}{100} = \frac{12}{5} = 2\frac{2}{5}$ hrs. = 2 hr. 24 min.

∴ Reqd. Difference = 3 hrs – 2 hrs. 24 min. = 36 mins.

22. 40 mins.;

Gauri's speed = 60 km/hr. and

Time taken by Gauri = 5 hrs.

∴ Distance covered by Gauri = $60 \times 5 = 300$ km.

Given, Distance covered by Pooja = Distance covered by Gauri = 300 km.

And time taken by Pooja = 4 hrs.

∴ Pooja's speed = $\frac{\text{Distance}}{\text{Time}} = \frac{300}{4} = 75$ km/hr.

Now, the average speed of Gauri and Pooja increases by 15 km/hr.

New speed of Gauri = $60 + 15 = 75$ km/hr.

New speed of Pooja = $75 + 15 = 90$ km/hr.

∴ Time taken by Gauri = $\frac{300}{75} = 4$ hrs. and

Time taken by Pooja = $\frac{300}{90} = \frac{10}{3} = 3\frac{1}{3}$ hrs. = 3 hr. 20 min.

∴ Reqd. Difference = 4 hrs – 3 hrs. 20 min. = 40 mins.

23. 5 : 7;

Car A:

Speed = 40 km/hr and time taken = 5 hrs.

∴ Distance = $\text{Speed} \times \text{Time} = 40 \times 5 = 200$ Km.

Car B:

Speed = 70 km/hr. and time taken = 4 hrs.

Distance = $\text{Speed} \times \text{Time} = 70 \times 4 = 280$ Km

Ratio of the distance covered by car A and car B = $200 : 280 = 5 : 7$

24. 24 : 35;

Car A:

Speed = 60 km/hr, Time = 4 hrs.

∴ Distance = $\text{Speed} \times \text{Time} = 60 \times 4 = 240$ Km.

Car B:

Speed = 50 km/hr, Time = 7 hrs.

∴ Distance = $\text{Speed} \times \text{Time} = 50 \times 7 = 350$ Km

Ratio of the distance covered by car A and car B = $240 : 350$

= 24 : 35

25. 35 m.;

Let the length of the train be x meter

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∴ When it crosses the pole, its Speed = $\frac{x}{5}$ m/sec
and

When it crosses the platform, its Speed = $\frac{70+x}{15}$ m/sec

$$\therefore \frac{x}{5} = \frac{70+x}{15} \Rightarrow 15x = 350 + 5x$$

$$\Rightarrow 10x = 350 \therefore x = 35 \text{ m.}$$

26. 60 m.;

Let the length of the train be x meter

∴ When it crosses the pole, its Speed = $\frac{x}{6}$ m/sec
and

When it crosses the platform, its Speed = $\frac{60+x}{12}$ m/sec

$$\therefore \frac{x}{6} = \frac{60+x}{12} \Rightarrow 12x = 360 + 6x$$

$$\Rightarrow 6x = 360 \therefore x = 60 \text{ m.}$$

27. 40 km / hr.;

$$\text{Average speed} = \frac{2xy}{x+y} = \frac{2 \times 60 \times 30}{60+30} = 40 \text{ km / hr.}$$

28. 42 km / hr;

$$\text{Average speed} = \frac{2xy}{x+y} = \frac{2 \times 70 \times 30}{70+30} = 42 \text{ km / hr.}$$

29. 100 km.;

Let distance travelled by Cycle = x km.

∴ distance travelled by Car = $(200 - x)$ km.

∴ Time taken by Cycle = $\frac{x}{30}$ hr. and

Time taken by Car = $\frac{200-x}{60}$ hr.

Given, Total time taken = 5 hrs.

$$\therefore \frac{x}{30} + \frac{200-x}{60} = 5 \Rightarrow \frac{2x + 200 - x}{60} = 5$$

$$\Rightarrow x + 200 = 300 \Rightarrow x = 100 \text{ km.}$$

∴ Distance travelled by Cycle = 100 km.

30. 70 km

Let distance travelled by Cycle = x km.

∴ distance travelled by Car = $(190 - x)$ km.

∴ Time taken by Cycle = $\frac{x}{30}$ hr. and

Time taken by Car = $\frac{190-x}{70}$ hr.

Given, Total time taken = 5 hrs.

$$\therefore \frac{x}{30} + \frac{190-x}{70} = 5 \Rightarrow \frac{7x + 570 - 3x}{210} = 5$$

$$\Rightarrow 4x + 570 = 1050 \Rightarrow 4x = 1050 - 570$$

$$\Rightarrow x = \frac{480}{4} \Rightarrow x = 120 \text{ km.}$$

∴ Distance travelled by Cycle = 120 km.

∴ Distance travelled by Car = $190 - 120 = 70$ km.

31. 3 hrs.;

$$\text{Time taken without halt} = \frac{\text{Distance}}{\text{speed}} = \frac{300}{60} = 5 \text{ hrs.}$$

∴ Time halted = $8 - 5 = 3$ hrs.

32. 2hr;

$$\text{Time taken without halt} = \frac{\text{Distance}}{\text{speed}} = \frac{500}{50} = 10 \text{ hrs.}$$

∴ Time halted = $12 - 10 = 2$ hr

33. 18 km.;

Let the distance between home and station be x km.

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$$\therefore \text{Total time taken} = \frac{x}{6} + \frac{x}{9} = 5$$

$$\Rightarrow \frac{3x + 2x}{18} = 5 \quad \Rightarrow 5x = 90$$

$$\therefore x = 18 \text{ km.}$$

34. 24 km.;

Let the distance between home and station be x km.

$$\therefore \text{Total time taken} = \frac{x}{4} + \frac{x}{6} = 10$$

$$\Rightarrow \frac{3x + 2x}{12} = 10 \quad \Rightarrow 5x = 120$$

$$\therefore x = 24 \text{ km.}$$

35. 24 km/hr.; Distance covered by the Bus = 2400 km.

Time taken by the Bus = 40 hrs.

$$\therefore \text{Speed of a bus} = \frac{\text{Distance}}{\text{Time}} = \frac{2400}{40} = 60 \text{ km/hr.}$$

$$\therefore \text{Average speed of the car} = \frac{2}{5} \times 60 = 24 \text{ km/hr.}$$

36. 10 km.; Distance covered by the train = 1200 km.

Time taken by the train = 15 hrs.

$$\therefore \text{Speed of the train} = \frac{\text{Distance}}{\text{Time}} = \frac{1200}{15} = 80 \text{ km/hr.}$$

$$\therefore \text{Average speed of a bus} = \frac{1}{4} \times 80 = 20 \text{ km/hr.}$$

$$\therefore \text{Distance covered by bus} = \text{Speed} \times \text{Time} = 20 \times \frac{30}{60} = 10 \text{ km.}$$

37. 80 km; Distance covered on bike = 300 km.

Time taken by the bike = 6 hrs.

$$\therefore \text{Speed of the bike} = \frac{\text{Distance}}{\text{Time}} = \frac{300}{6} = 50 \text{ km/hr}$$

$$\therefore \text{Speed of a bicycle} = \frac{2}{5} \times 50 = 20 \text{ km/hr.}$$

$$\therefore \text{Distance covered by bicycle} = 20 \times 4 = 80 \text{ km}$$

38. 40 km/hr.;

$$\text{Total Distance covered} = 60 \times 4 = 240 \text{ km.}$$

$$\therefore \text{Speed should be to cover the same distance} = \frac{240}{6} =$$

$$40 \text{ km/hr.}$$

39. 7 m/s; Total Distance = $D_1 + D_2 + D_3 = 108 + 110 + 160 = 378 \text{ km.}$

$$\text{Total time taken} = \frac{D_1}{S_1} + \frac{D_2}{S_2} + \frac{D_3}{S_3} = \frac{108}{108} + \frac{110}{11} + \frac{160}{40}$$

$$= 1 + 10 + 4 = 15 \text{ hrs.}$$

$$\therefore \text{Average Speed} = \frac{\text{Total distance covered}}{\text{Total time taken}} = \frac{378}{15} \times \frac{5}{18} = 7 \text{ m/s}$$

40. 300 km/hr.;

Let the speed of the car be x

$$\therefore \text{Speed of the train} = 60 + x \text{ km/hr.}$$

$$\text{Given, } \frac{x}{60 + x} = \frac{10}{12} \quad \Rightarrow 12x = 600 + 10x$$

$$\Rightarrow 2x = 600 \quad \therefore x = 300 \text{ km/hr.}$$

41. 105 km/hr; Let the speed of the car be x

$$\therefore \text{Speed of the train} = 30 + x \text{ km/hr.}$$

$$\text{Given, } \frac{x}{30 + x} = \frac{7}{9} \quad \Rightarrow 9x = 210 + 7x$$

$$\Rightarrow 2x = 210 \quad \therefore x = 105 \text{ km/hr.}$$

42. 20 minutes;

$$\text{Speed of train A} = \frac{40}{25} \times 60 = 96 \text{ km/hr}$$

$$\therefore \text{Speed of train B} = 96 \times \frac{125}{100} = 120 \text{ km/hr}$$

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∴ Time taken by train B = $\frac{40}{120} = \frac{1}{3}$ hr. = $\frac{1}{3} \times 60 = 20$ minutes

43. **90 km.;**

Let the distance be x km.

∴ Time taken at 6 km/hr = $\frac{x}{6}$ hr. and

Time taken at 9 km/hr = $\frac{x}{9}$ hr.

Difference in time taken at different speeds =

= 2 hrs 30 minutes + 2 hrs 30 minutes = 5 hrs.

∴ $\frac{x}{6} - \frac{x}{9} = 5 \Rightarrow \frac{x}{18} = 5 \Rightarrow x = 90$ km.

44. **72 km.;** Let the distance be x km.

∴ Time taken at 8 km/hr = $\frac{x}{8}$ hr. and

Time taken at 9 km/hr = $\frac{x}{9}$ hr.

Difference in time taken at different speeds =

= 20 minutes + 40 minutes = 1 hr

∴ $\frac{x}{8} - \frac{x}{9} = 1 \Rightarrow \frac{x}{72} = 1 \Rightarrow x = 72$ km.

45. **770 km;** Distance covered by the tractor = 495 km.

Time taken by the tractor = 15 hrs.

∴ Speed of the tractor = $\frac{\text{Distance}}{\text{Time}} = \frac{495}{15} = 33$ km/hr

∴ Speed of a train = $\frac{10}{3} \times 33 = 110$ km/hr.

∴ Distance covered by the train = $110 \times 7 = 770$ km

46. **756 km.;** Speed of the car = $\frac{720}{9} = 80$ km/hr. and

∴ Speed of the bus = $\frac{3}{4} \times 80 = 60$ km/hr.

According to the question, $15x = 60 \Rightarrow x = 4$ km/hr.

∴ Speed of a train = $27 \times 4 = 108$ km/hr

∴ Distance covered in 7 hrs. = $108 \times 7 = 756$ km.

47. **450 km.;** Let the distance covered by him each way be x km.

∴ $\frac{x}{45} - \frac{x}{50} = 1 \Rightarrow \frac{10x - 9x}{450} = 1$

∴ $x = 450$ km.

48. **52 km/hr.;**

Required average speed = $\frac{\text{Total dist.covered}}{\text{Total time taken}}$

= $\frac{104 \times 60}{120} = 52$ km/hr.

49. **65 km/hr;** Let total distance be x km.

Dist. Covered by the train in $7\frac{1}{2}$ hrs. = $40 \times 3 + 60 \times 4.5 = 390$ km.

∴ $\frac{3x}{5} = 390 \Rightarrow 3x = 390 \times 5$

∴ $x = \frac{390 \times 5}{3} = 650$ km.

∴ Remaining distance = $650 - 390 = 260$ km.

∴ Required average speed = $\frac{260}{4} = 65$ km/hr.

50. **3:4;**

Time taken by Car = 2 hr. 40 min. = $2\frac{2}{3}$ hr. = $\frac{8}{3}$ hr.

∴ Speed of the car = $\frac{200 \times 3}{8} = \frac{600}{8} = 75$ km/hr and

Speed of the Jeep = $\frac{200}{2} = 100$ km/hr

∴ Required ratio = $75 : 100 = 3:4$